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Aspartame Consumption for 12 Weeks Does Not Affect Glycemia, Appetite, or Body Weight of Healthy, Lean Adults in a Randomized Controlled Trial

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Abstract

Background: Low-calorie sweeteners are often used to moderate energy intake and postprandial glycemia, but some evidence indicates that they may exacerbate these aims.

Objective: The trial's primary aim was to assess the effect of daily aspartame ingestion for 12 wk on glycemia. Effects on appetite and body weight were secondary aims.

Methods: One hundred lean [body mass index (kg/m^2): 18–25] adults aged 18–60 y were randomly assigned to consume 0, 350, or 1050 mg aspartame/d (ASP groups) in a beverage for 12 wk in a parallel-arm design. At baseline, body weight and composition were determined, a 240-min oral-glucose-tolerance test (OGTT) was administered, and measurements were made of appetite and selected hormones. Participants also collected a 24-h urine sample. During the intervention, the 0-mg/d ASP group consumed capsules containing 680 mg dextrose and 80 mg para-amino benzoic acid. For the 350-mg/d ASP group, the beverage contained 350 mg aspartame and the 1050-mg/d ASP group consumed the same beverage plus capsules containing 680 mg dextrose and 700 mg aspartame. Body weight, blood pressure, heart rate, and waist circumference were measured weekly. At weeks 4, 8, and 12, participants collected 24-h urine samples and kept appetite logs. Baseline measurements were repeated at week 12.

Results: With the exception of the baseline OGTT glucose concentration at 60 min (and resulting area under the curve value), there were no group differences for glucose, insulin, resting leptin, glucagon-like peptide 1, or gastric inhibitory peptide at baseline or week 12. There also were no effects of aspartame ingestion on appetite, body weight, or body composition. Compliance with the beverage intervention was ~95%.

Conclusions: Aspartame ingested at 2 doses for 12 wk had no effect on glycemia, appetite, or body weight among healthy, lean adults. These data do not support the view that aspartame is problematic for the management of glycemia, appetite, or body weight. This trial was registered at www.clinicaltrials.gov as NCT02999321. *J Nutr* 2018;148:650–657.

Keywords: aspartame, glycemia, body weight, appetite, low calorie sweetener

Introduction

The high global prevalence of overweight and obesity in children and adults is well documented (1). One of the many complications associated with high adiposity is impaired glucose tolerance and diabetes (2, 3). Low-calorie sweeteners (LCSs) are added to foods and beverages by the food industry and consumers for multiple economic, sensory, and health purposes. A key driver for LCS use is to displace sugar intake and thereby reduce total energy intake and the glycemic response to sweet food ingestion. LCS consumption by children and adults is high and increasing (1, 4–7). Approximately 25% of children and 41% of adults report using these products (7) and many more consume them unwittingly, because they have become pervasive in the food supply (8).

Recent large meta-analyses indicate that the use of LCSs is associated with lower BMI (9, 10), and LCSs are commonly used by individuals with sustained weight loss in the National Weight Control Registry (11). Several professional societies, including the American Heart Association (12), the Academy of Nutrition and Dietetics (13), and the Canadian Diabetes Association (14), have noted that LCSs either moderate [e.g., (15, 16)] or have no effect on postprandial glycemia, but none has concluded that they exacerbate blood sugar excursions. Their use has also been associated with a higher Healthy Eating Index (17). With this evidence base, >25% of consumers report that they would pay a premium price for sugar-free products (18). However, some researchers maintain that LCSs undermine attempts to manage body weight and glycemia (19–29). Thus, the issue remains contentious.

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Questions related to the safety of LCSs have been extensively reviewed by the US FDA and multiple international bodies (30). These products have consistently been deemed safe by these agencies, although this conclusion is still challenged by some investigators [e.g., (26)]. This issue aside, mechanisms that have been posited for the adverse effects of LCSs on body weight, glycemia, or both include "hijacking" of sweet receptors with resulting hyperstimulation of brain reward systems; activation of a cephalic phase insulin response; disassociation of sweetness and energy yield, training the palate to prefer very sweet foods and beverages; increased hunger and desire to eat; and enhanced energy absorption from the gastrointestinal tract via augmented glucose transport or shifts in gut microflora leading to more efficient energy harvesting. Evidence that LCSs are associated with lower BMI (9, 10) and do not promote hyperglycemia (15, 31) notwithstanding, the plausibility of these mechanisms still provides grounds for questioning the health effects of LCSs.

Aspartame is one widely used LCS implicated in the claim that LCSs are problematic with respect to energy balance and glycemia (21, 29). However, there is reason to question this view, because each of the proposed mechanisms is of uncertain validity. First, aspartame and other LCSs do not elicit greater sweetness intensity than sucrose (32). They may achieve their maximal effect at a lower concentration than sucrose, but they do not elicit greater perceived sweetness and may not be as strong a stimulus for dopamine release as nutritive sweeteners (15, 33, 34). Second, several investigators report that oral sweet exposure is not a sufficient stimulus for insulin release (35–37). Although there is documentation that some LCSs have supported cephalic phase insulin responses (38) [or biphasic responses in β cells; acesulfame potassium (ace-K) (39)], aspartame has not proven to be an effective stimulus for insulin release (40, 41). Such a lack of effect could be argued equally to lead to a reduced, instead of a larger, meal size (42). Other work indicates that oral exposure to an LCS (e.g., stevioside, sucralose) reduces rather than increases postprandial glycemia (43, 44). Third, there is very limited evidence of a hedonic training effect. In one trial, adults exposed to intensely sweet foods showed no hedonic shift (45), although a very small effect was noted in children. The intervention entailed added sucrose so it was reinforcing through both its sensory and energy-yielding properties. Aspartame would contribute negligible energy, and energy may play an underappreciated role in supporting a learned preference effect (46, 47). In other work, no difference in food choice, implicit intake, explicit liking, or expected satiety was observed after the consumption of soft drinks and yogurt drinks containing a nutritive sweetener and LCSs (48). Furthermore, even if aspartame enhanced the appeal of highly sweetened foods, this is not a reliable predictor of food choice or energy intake in individuals who are lean or who have high adiposity (49). Fourth, with respect to augmentation of appetitive sensations, the preponderance of the literature fails to support this effect [e.g., (50, 51)]. Fifth, a mechanism involving enhanced glucose transport has been supported by cell culture studies (52) but not in vivo human trials (53–55). Moreover, even if the time course of glucose absorption was altered by oral aspartame exposure, an effect on energy balance is doubtful because the efficiency of glucose absorption is extremely high. Thus, there would be a ceiling effect. Finally, because aspartame is a dipeptide, it is efficiently hydrolyzed in the small intestine so would not be expected to reach the colon and exert an effect on the local microbiome.

Given the ongoing controversy and lack of a likely mechanism, a larger, longer-term, and more definitive study of aspartame's effect on postprandial glycemia was undertaken. Acute effects were assessed as well as the effect of daily consumption of 3 doses (0, 350, and 1050 mg/d) for 3 mo by healthy adults with a normal BMI. The primary outcome was the glycemic response. Secondary outcomes were appetite, body weight, and body composition.

Methods

Study design. This was a randomized, 3-parallel-arm design trial. Participant assignment was sequential based on order of recruitment with no adjustments. This trial was registered at www.clinicaltrials.gov as NCT02999321.

Participants. A total of 214 individuals responded to advertisements for the study, and 100 were randomly assigned to treatments between 10 August 2016 and 7 February 2017 (**Figure 1**). During a prescreening assessment, participants completed a battery of questionnaires to obtain baseline health and demographic information and had measurements taken of body weight, waist circumference, blood pressure, and heart rate. Those meeting eligibility criteria and who chose to participate signed a consent form approved by the Purdue University Institutional Review Board.

The eligibility criteria were as follows: male or female aged 18–60 y, BMI (kg/m^2) of 18–25, not taking medications that affect metabolism or appetite, fasting serum glucose between 4.0 and 6.0 mmol/L (72–108 mg/dL) via capillary finger-stick, willing to comply with the study protocol, no reported aspartame sensitivity, and non- or low user of LCSs, including aspartame. The latter was determined during the prescreening assessment in which individuals reported low (<1 time/wk) or no discretionary use of LCSs and no purposeful selection of products containing LCSs. They also indicated nonuse of items on an FFQ composed of commercial products containing LCSs that are reduced in energy density by >50% through substitution of an LCS for a nutritive sweetener.

Participants were sequentially randomly assigned on the basis of a generated random order (https://www.randomizer.org/) to 1 of 3 groups, resulting in samples of 33 in the group ingesting no aspartame, 33 in the group ingesting 350 mg aspartame/d, and 34 in the group ingesting 1050 mg aspartame/d. A total of 93 individuals completed the trial, 31 in each of the groups. Thus, attrition was 2, 2, and 3 in the groups ingesting 0, 350, or 1050 mg aspartame/d, respectively. Withdrawals were attributable to conflicts of time management or loss of interest but not because of known study-related complications (i.e., adverse side effects).

Protocol. At baseline, participants were instructed on methods for 24-h urine collection and for recording appetitive sensations over the waking part of 24-h periods. Appetite ratings were self-reported on smartphones via Qualtrics software under free-living conditions. Participants were instructed to record their sensation for each

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Supplemental Figures 1 and 2 are available from the "Supplementary data" link in the online posting of the article and from the same link in the online table of contents at https://academic.oup.com/jn/.
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Abbreviations used: ace-K, acesulfame potassium; ASP group, aspartame group; GIP, gastric inhibitory peptide; GLP-1, glucagon-like peptide 1; HbA1c, glycated hemoglobin; LCS, low-calorie sweetener; OGTT, oral glucose tolerance test; PABA, para-amino benzoic acid.

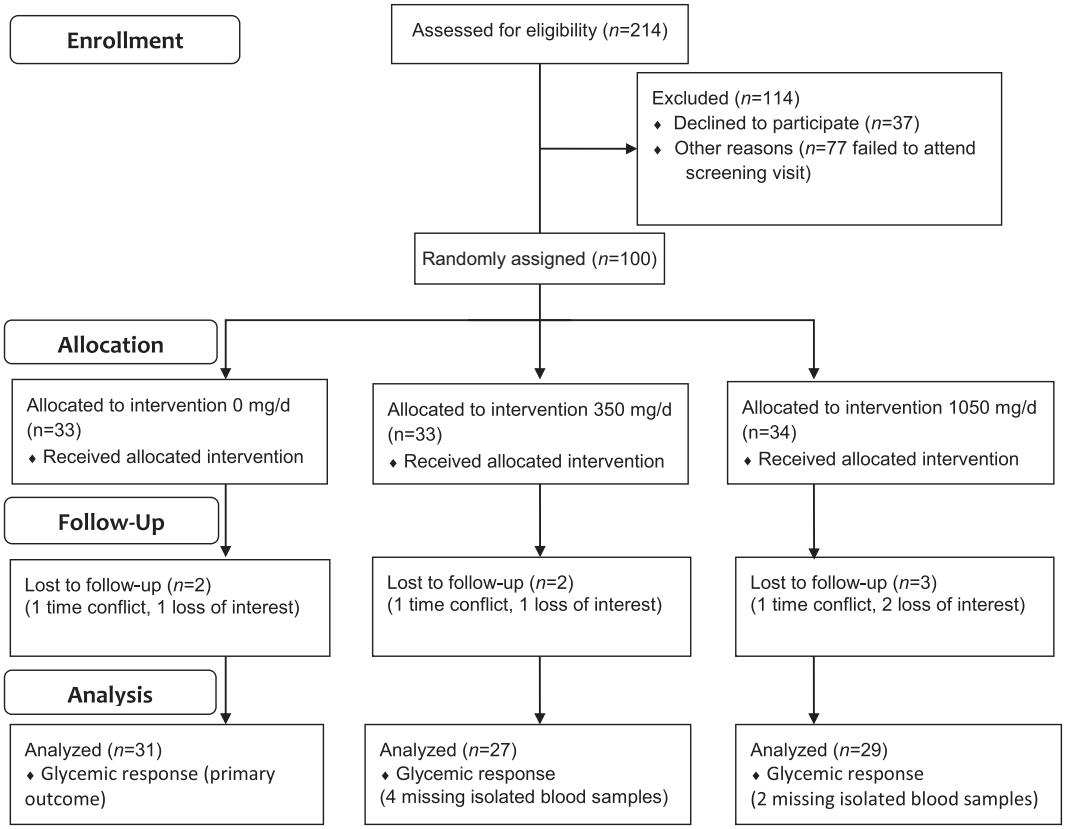


FIGURE 1 Participant flow chart.

quality (hunger, fullness, desire to eat, thirst, prospective consumption, preoccupation with food) on an hourly basis for each waking hour of the day. After completion of the urine collection and appetite log, participants reported to the laboratory in a fasted state. An indwelling catheter was placed in an arm vein, and a baseline blood sample was obtained 10 min after catheter insertion. A blood sample was collected for assessment of fasting glucose, glycated hemoglobin (HbA1c), the incretin hormones gastric inhibitory peptide (GIP) and glucagon-like peptide 1 (GLP-1), insulin, leptin, total cholesterol, LDL cholesterol, HDL cholesterol, TGs, γ-glutamyltranspeptidase, alanine transaminase, and aspartate transaminase. Next, participants ingested a 296-mL bottle of glucola (75 g glucose) within 10 min and additional blood draws were taken at 15, 30, 60, 90, 120, 180, and 240 min. This was followed by a body-composition measurement via plethysmography (Bod Pod, COSMED).

During the intervention phase, participants consumed an assigned beverage and capsules daily for 12 wk. They reported to the laboratory weekly to pick up their next week's supply of beverage powder and capsules; measurement of body weight, blood pressure, heart rate, and waist circumference; as well as assessment of study progress. At weeks 4 and 8, they collected a 24-h urine sample and kept a 24-h appetite log. At the end of 12 wk, participants repeated the baseline measurements.

Intervention. Active treatment participants consumed 500 mL of fruit-flavored beverages sweetened with aspartame (preferred flavors self-selected from 3 alternatives) and 4 capsules each day. This provided 80 mg para-amino benzoic acid (PABA), 680 mg dextrose, and either 350 mg or 1050 mg aspartame (the equivalent of 0, 5, or 15 mg aspartame/kg based on a 70-kg reference individual). The specific group interventions were as follows:

- 0-mg/d aspartame group (ASP group): 2 capsules collectively containing 680 mg dextrose and 80 mg PABA, and 2 empty capsules
- 350-mg/d ASP group: premeasured sachets of lemon-, orange-, or fruit punch–flavored dry powder beverage mixture reconstituted by participants to yield 500 mL containing 350 mg aspartame and 80 mg PABA; 2 capsules collectively containing 680 mg dextrose and 2 empty capsules
- 1050-mg/d ASP group: premeasured sachets of lemon-, orange-, or fruit punch–flavored dry powder beverage mixture reconstituted by participants to yield 500 mL containing 350 mg aspartame and 80 mg PABA; 4 capsules collectively containing 700 mg aspartame and 680 mg dextrose

Analyte assays. Blood samples were drawn from a catheter into evacuated tubes for preparation of serum or EDTA plasma. Dipeptidyl peptidase-4 inhibitor (Millipore Corporation) was immediately added to the plasma tube; samples were centrifuged at 4°C at 3000 × *g* for 10 min, separated into aliquots, and frozen in a –80°C freezer within 15 min of the draw. Serum insulin, glucose, cholesterol (total, HDL, LDL), TGs, γ-glutamyltranspeptidase, alanine transaminase, and aspartate transaminase were measured on a Roche COBAS 400 Plus analyzer. HbA1c was determined in whole blood with the use of the COBAS analyzer. GLP-1 active (7–36) and GIP were quantitated in plasma by using ELISA (Millipore) with minimum sensitivities of 0.14 pM and 4.2 pg/mL, respectively. All of the samples for a participant (pre- and postintervention) were measured on the same ELISA plate. Serum leptin was measured by using an RIA (Millipore). All of the samples for an individual were measured in the same assay.

Compliance. PABA was inextricably mixed with the beverage powder. Analysis of 24-h urine samples for PABA recovery followed the method of Sharma et al. (56). Urine samples were collected at baseline to establish the PABA composition of each individual's customary diet and again at weeks 4, 8, and 12. Creatinine was measured on a COBAS 400 Plus analyzer to ensure complete 24-h samples were collected and, if not, then corrections were applied as previously published (57).

Statistical analysis. A mixed-model repeated-measures ANOVA was used to assess glucose and insulin responses, appetite, body weight, and body composition. Session (baseline, end of week 12) and time (0–240 min) were within-subjects factors and treatment group was the

TABLE 1 Baseline characteristics of study participants enrolled in the randomized controlled trial[1]

ASP group	Sex (M/F), *n*/*n*	Age, y	BMI, kg/m^2	WC, cm	SBP, mm Hg	DBP, mm Hg	HbA1c (whole blood), %	Fasting serum glucose, mg/dL
0 mg/d	13/18	24.2 ± 1.3	21.7 ± 0.3	74.1 ± 1.3	118.8 ± 2.1	73.4 ± 1.4	5.13 ± 0.04	83.5 ± 1.4
350 mg/d	17/14	22.8 ± 1.0	22.3 ± 0.3	74.5 ± 1.1	123.8 ± 2.4	76.1 ± 1.7	5.12 ± 0.04	81.5 ± 1.5
1050 mg/d	13/18	21.8 ± 0.6	22.1 ± 0.3	71.4 ± 2.9	124.0 ± 2.8	77.2 ± 1.4	5.15 ± 0.04	81.6 ± 1.5

[1]Values are means ± SDs, $n = 31$/group. There were no significant differences between treatment groups for for age, BMI, waist circumference, systolic or diastolic blood pressure, HbA1c, or fasting glucose at baseline. ASP group, aspartame group; DBP, diastolic blood pressure; HbA1c, glycated hemoglobin; SBP, systolic blood pressure; WC , waist circumference.

between-subjects factor. Analyses were conducted by using the IBM SPSS Statistics software package (version 23). The α level was set at $P < 0.05$ (2-tailed). Bonferroni correction was used for post hoc analysis.

Results

Baseline characteristics. There were no differences in baseline characteristics between groups on the basis of the total recruited and among completers. Baseline characteristics of those completing the trial are presented in Table 1. There were no significant differences between treatment groups for age, BMI, waist circumference, systolic or diastolic blood pressure, HbA1c, or fasting glucose at baseline.

Biochemical outcomes. The effects of daily aspartame consumption over 12 wk on the hematologic indexes monitored are presented in Table 2. There was no significant main effect of treatment or session, or a treatment by session interaction for HbA1c, lipids, or liver function indexes. The estimated compliance with beverage consumption on the basis of PABA and creatinine excretion was 95.4% ± 11.8% in aspartame-consuming treatment groups.

TABLE 2 HbA1c and serum biochemistry of participants at baseline and after 12 wk of daily consumption of 0, 350, or 1050 mg aspartame in a beverage[1]

	0 mg/d	350 mg/d	1050 mg/d
HbA1c, %			
Baseline	5.13 ± 0.04	5.12 ± 0.04	5.15 ± 0.04
Week 12	5.20 ± 0.04	5.22 ± 0.04	5.23 ± 0.04
Total cholesterol, mg/dL			
Baseline	157.2 ± 4.8	149.4 ± 5.0	148.7 ± 5.0
Week 12	158.4 ± 4.6	151.6 ± 4.8	153.7 ± 4.8
HDL cholesterol, mg/dL			
Baseline	53.6 ± 2.3	54.0 ± 2.4	59.0 ± 2.4
Week 12	56.7 ± 2.5	56.7 ± 2.6	61.0 ± 2.6
LDL cholesterol, mg/dL			
Baseline	88.5 ± 4.2	79.3 ± 4.4	72.7 ± 4.4
Week 12	85.8 ± 3.8	78.6 ± 4.0	76.2 ± 4.0
TGs, mg/dL			
Baseline	75.7 ± 6.1	79.4 ± 6.3	85.1 ± 6.3
Week 12	80.4 ± 6.4	81.4 ± 6.6	85.5 ± 6.6
GGTP, U/L			
Baseline	14.2 ± 1.5	12.9 ± 1.5	13.3 ± 1.6
Week 12	14.9 ± 1.6	13.9 ± 1.7	15.0 ± 1.7
ALT, U/L			
Baseline	13.3 ± 1.3	13.5 ± 1.3	12.7 ± 1.3
Week 12	14.5 ± 1.6	14.3 ± 1.6	16.4 ± 1.7
AST, U/L			
Baseline	19.6 ± 1.5	19.5 ± 1.6	19.1 ± 1.6
Week 12	20.6 ± 1.9	21.2 ± 2.0	19.8 ± 2.0

[1]Values are means ± SEs, $n = 31$/group. Variables did not differ between groups at baseline or change over time. ALT, alanine transaminase; AST, aspartate transaminase; GGTP, γ-glutamyltranspeptidase; HbA1c, glycated hemoglobin.

Baseline and week 12 serum glucose values from the oral-glucose-tolerance tests (OGTTs) are plotted in Figure 2A, B, respectively. Data are reported on 27 and 30 participants in the groups ingesting 350 and 1050 mg aspartame/d, respectively, due to complications in sample collection. There was no significant main effect of treatment or session. There was a significant time (0–240 min) by treatment (0, 350, or 1050 mg/d) by session (baseline, week 12) interaction ($P < 0.03$). The 350-mg/d ASP group had lower serum glucose concentrations at 60 min than did the 0-mg/d ($P < 0.01$) or the 1050-mg/d ($P < 0.05$) ASP groups at baseline, but there were no significant treatment group differences at week 12. Overall glucose values were significantly higher at week 12 compared with baseline ($P < 0.05$). Baseline and week 12 insulin values from the OGTT tests are plotted in Figure 2C, D, respectively. There was no main effect of treatment, nor was there a significant treatment by session interaction.

GLP-1 and GIP concentrations during the OGTT at baseline and week 12 are presented in **Supplemental Figure 1**. No significant main effect of treatment or treatment by session interaction was observed for either of the incretins. Resting leptin concentrations also did not differ by treatment or over time (**Supplemental Figure 2**).

Appetite. Ratings of hunger, desire to eat, fullness, and thirst are presented in Figure 3. Values represent the mean of the totality of data available for each group and time point (i.e., *n* values differed). Because of differences in the timing and duration of each person's waking time and numerous instances in which hourly ratings were missed, the analysis of changes over time was restricted to individuals who captured ≥6 hourly ratings at baseline and week 12 ($n = 16$, 12, and 16 for the 0-, 350-, and 1050-mg/d ASP groups). No significant differences were observed for hunger, desire to eat, fullness, or thirst in this subset of appetite ratings. These ratings did not differ from the smaller subset ($n = 5$, 6, 9) who recorded ≥12 hourly ratings each day.

Body weight. Participants were not placed on an energy-restricted diet. The only guidance provided was that they consume the daily beverage portion, which contained no, or a trivial amount of, energy. Thus, no weight change was expected in any treatment group. There was no main effect of treatment or a treatment by session interaction for body weight. There also were no significant main effects of treatment or session or an interaction between treatment and session for percentage of fat mass, percentage of fat-free mass, fat mass, or fat-free mass (Table 3).

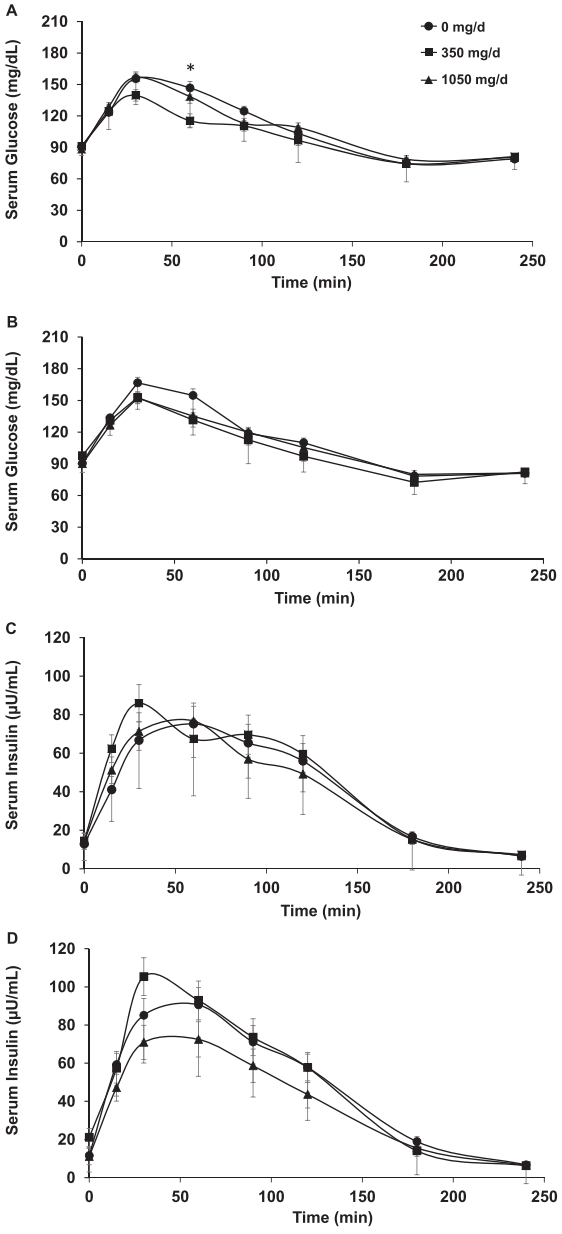


FIGURE 2 Serum glucose (A, B) and insulin (C, D) concentrations during the oral-glucose-tolerance test for participants consuming 0 mg ($n = 31$), 350 mg ($n = 27$), or 1050 mg ($n = 30$) aspartame/d in a beverage at baseline (A, C) and after 12 wk (B, D) of daily consumption. Values are means ± SEs. *Different between 0- and 350-mg/d treatments, $P < 0.05$. There were no significant differences between 0- and 1050-mg/d treatments or 350- and 1050-mg/d treatments.

Discussion

The primary outcome of this trial was the glycemic response to chronic aspartame ingestion. The 350-mg/d dose corresponds to an intake of ~1 can of low-calorie soda/d and the mean of intake for aspartame consumers, whereas the 1050-mg/d dose approximates the 95th percentile for consumers (58). The glycemic response to each treatment (including a no-aspartame control) was measured at baseline and again after 12 wk of daily beverage ingestion. The data show that responses after the ingestion of the 350- or 1050-mg/d doses were not significantly higher than the no-aspartame treatment either at baseline or after the 12-wk daily consumption period. The 1 random single time point difference is not viewed as meaningful.

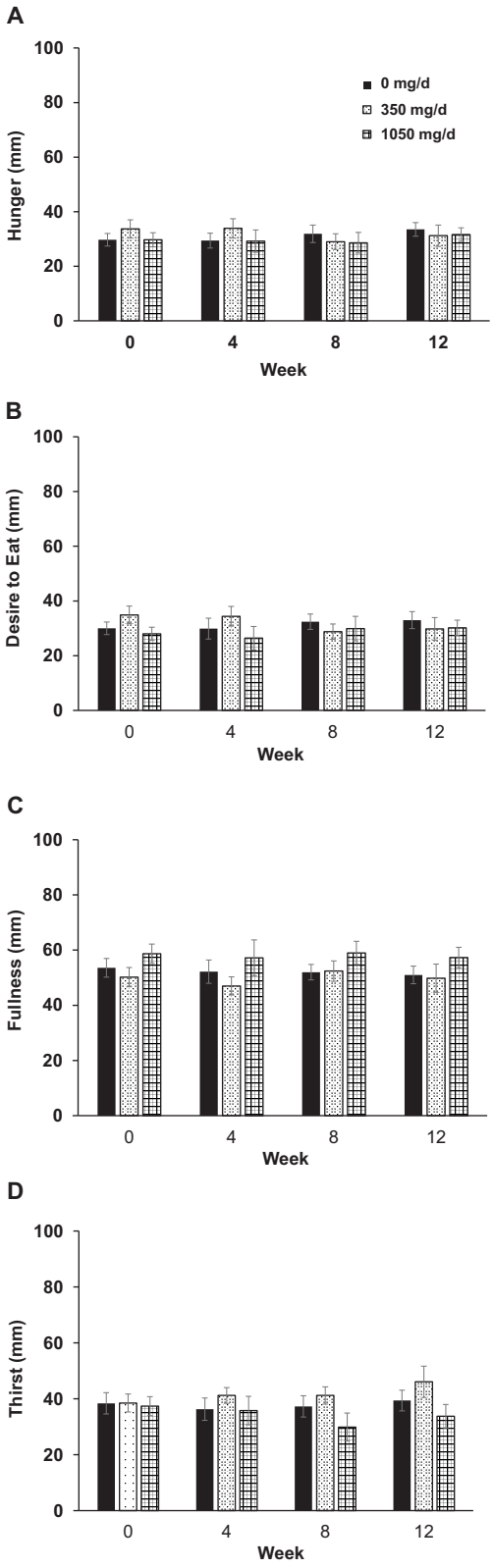


FIGURE 3 Appetite ratings for hunger (A), desire to eat (B), fullness (C), and thirst (D) by participants consuming 0 mg ($n = 16$), 350 mg ($n = 12$), or 1050 mg ($n = 16$) aspartame/d in a beverage at baseline and after 12 wk of daily consumption. Values are means ± SEs. No significant differences were observed for hunger, desire to eat, fullness, or thirst.

TABLE 3 Body-composition indexes of participants at baseline and after consumption of 0, 350, or 1050 mg aspartame/d in a beverage for 12 wk[1]

	Body weight, kg		Fat mass, %		Fat-free mass, %		Fat mass, kg		Fat-free mass, kg	
ASP group	Baseline	Week 12	Baseline	Week 12	Baseline	Week 12	Baseline	Week 12	Baseline	Week 12
0 mg/d	63.5 ± 1.8	63.0 ± 1.8	22.8 ± 1.6	22.6 ± 1.6	75.2 ± 2.0	77.4 ± 1.6	14.1 ± 1.6	13.7 ± 0.9	49.0 ± 2.0	49.2 ± 2.1
350 mg/d	65.3 ± 1.8	65.6 ± 1.8	19.6 ± 1.6	21.0 ± 1.6	79.4 ± 2.0	79.0 ± 1.6	13.3 ± 1.6	13.7 ± 0.9	52.3 ± 2.1	52.6 ± 2.2
1050 mg/d	63.6 ± 1.8	63.8 + 1.8	22.4 ± 1.6	21.8 ± 1.6	77.5 ± 2.0	78.2 ± 1.6	14.0 ± 1.6	13.5 ± 0.9	49.7 ± 2.0	50.3 ± 2.1

[1]Values are means ± SEs, $n = 31$/group. There were no significant differences between treatment groups or over time. ASP group, aspartame group.

A true lack of effect is supported by urinary biomarker excretion findings of ~95% compliance with the aspartame ingestion intervention. Similarly, the insulin, GLP-1, and GIP responses after both doses of aspartame were not significantly different from the control condition. Body composition and fasting leptin were also unaffected by aspartame ingestion.

These findings confirm and extend the current understanding of the effects of aspartame and other LCSs on glycemia. Early interventional studies indicated that LCSs did not alter the glycemic response in healthy individuals, individuals with obesity, or patients with type 1 or type 2 diabetes (15, 31, 59, 60). However, with the discovery of sweet receptors in the gastrointestinal tract, new interest arose with regard to the implications of sweetener signaling on incretin secretion, glucose absorption, and peripheral glycemia (61). However, neither increased incretin secretion nor glucose absorption were observed in response to aspartame in in vitro trials with endocrine and absorptive cells or in in vivo animal or human trials. The same cannot be said for other LCSs. Cell culture work suggested that sucralose enhances GLP-1 secretion (62) and a variety of LCSs (except for aspartame) activate the glucose transporters sodium-glucose linked transporter 1 and glucose transporter 2 and glucose absorption (52, 63).

Enhanced absorption of glucose was observed in animal trials with exposure to ace-K, saccharin, and sucralose (52), but this was not supported by other studies involving an array of LCSs (64). An early trial in humans noted that the consumption of Diet Rite Soda sweetened with sucralose and ace-K before an OGTT led to a greater increase in GLP-1 than did carbonated water, although glucose and insulin concentrations were unaffected (53). To isolate sweetener effects in the gastrointestinal tract, sucralose was infused into the duodenum, with no effect on incretin secretion, glucose absorption, or glycemia (54); and aspartame, ace-K, and sucralose were administered intragastrically without effect on gut peptides (65). In contrast, effects were noted with nutritive sweeteners. The preponderance [not totality (66)] of human trials has since confirmed the lack of LCS on incretin release and either no effect, or a lowering of glycemia, insulinemia, or both (15, 44, 55, 67, 68). Interestingly, several trials report an augmentation by diet cola (6, 31, 53, 67), but the effect appears to be related to the cola rather than to the sweetener (6). The present findings with a sample size 3–9 times greater than previous similar trials are consistent with this body of evidence and extend the understanding by documenting that daily aspartame consumption for 12 wk has no effect on incretin release, glycemia, or insulinemia.

An a priori–identified secondary outcome pursued in this work related to the effects of aspartame on appetite. Some early work suggested that oral exposure to aspartame in a medium that did not provide energy (e.g., water or chewing gum) could augment hunger (51, 69–71). However, others did not replicate these findings (69, 72, 73), and subsequent reviews and later clinical trials of LCSs, including aspartame, indicate that LCSs do not augment hunger (15, 60, 65, 74–82). One group reported that the ingestion of aspartame in a capsule decreased hunger (70, 83, 84), although this finding has not been replicated. We observed no effect of aspartame exposure at either the moderate or high dose relative to the no-aspartame control at any time point and no shift in hunger, fullness, or desire-to-eat ratings over time in any treatment group. Participants were free-living and not instructed as to when and how to consume their daily beverage. In addition, acute effects were not monitored, precluding a direct test of hypotheses related to either hunger stimulation or suppression after single ingestive events. Nevertheless, it may be noted that despite many participants indicating that they frequently consumed the non–energy-yielding beverages alone, no augmentation of hunger was detected and in the 1050-mg/d ASP group, where two-thirds of the dose was delivered in capsules, no reduction in hunger was detected. Our appetite findings with normal-weight individuals are also consistent with a 10-wk trial in individuals who were overweight and consumed a beverage with aspartame as the primary sweetener (85).

The effects of aspartame on body composition were an additional secondary outcome. Recent meta-analyses have yielded conflicting conclusions about the role of LCSs (type not segregated) on BMI. In 2 reports, cohort studies showed that LCS use is associated with higher (9, 86) or no change (10) in BMI, whereas randomized controlled trials indicate that they are associated with lower (9) or no effect (86) on BMI. Both cohort studies and randomized controlled trials indicated that reported energy intake is lower with LCS use (10, 87), except where comparisons are made to treatments with water, nothing, or unsweetened products. In the latter cases, findings are that LCSs led to lower (87) or comparable (10) effects on energy intake. This seeming inconsistency likely reflects multiple subtle analytical differences. Among these are differences in study inclusion criteria (all trials base their analysis on different studies; some published in close temporal contiguity actually have no overlap in trials), the nature of the comparisons made (LCSs compared with different treatments), dissimilar study participants (lean, obese, minors, adults, etc.), duration of study, sample size, and types of LCSs. Resolution of the issue will likely require additional data to build the database to a point where there is adequate power to tease out these nuances.

In summary, the present trial found that daily ingestion of aspartame for 12 wk in a fruit-flavored beverage did not elicit any change in glycemia, incretin release, appetite, body weight, or body composition in healthy, lean adults. The present study was not intended to evaluate aspartame's efficacy for purposeful weight loss, but the findings do not indicate that its use as a substitute for energy-yielding sweeteners would be counterproductive.

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